

Lesson 31: Polar Coordinates

Polar Coordinates Assessment Rubric

Criteria	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Plotting Points	Plots all points (r, θ) accurately, including negative r or θ .	Plots most points accurately, minor errors with negative values.	Struggles with angle orientation or negative radius plotting.	Cannot locate points on the polar grid.
Conversion (P $\rightarrow$ R)	Flawlessly converts between (r, θ) and (x, y) using $x = r \cos \theta$, $y = r \sin \theta$, $r^2 = x^2 + y^2$.	Converts accurately with minor calculator or algebraic errors.	Understands formulas but mixes up sin/cos or quadrant signs.	Unable to apply conversion formulas.
Graphing Equations	Accurately identifies and graphs polar shapes (cardioids, roses) with correct scaling and orientation.	Graphs shapes correctly but with minor errors in size, period, or scaling.	Attempts to graph but misses symmetry or key features of the polar curve.	Graphs are unrecognizable or plotted as Cartesian.
Equivalent Points	Identifies all equivalent pairs, including $(r, \theta + 2n\pi)$ and $(-r, \theta + (2n+1)\pi)$.	Identifies 1-2 alternative representations correctly.	Finds equivalent points only for simple cases (adding 2π).	Cannot find equivalent representations.
Conversion (Equations)	Converts complex polar equations to Cartesian (x, y) and vice versa efficiently.	Converts basic equations accurately (e.g., $r = 2 \sin \theta$).	Makes mistakes in algebraic manipulation or trigonometric substitution.	Unable to convert equations.